



Privacy-Preserving “Federated” Symptom Checker

A Research Proposal Synopsis

(Based on Detailed Literature Review)

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Technology

in Computer Science and Engineering

Session: 2023–2027

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August 20, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

We also declare that the intellectual content of this synopsis is the product of our own work, even though we may have received assistance from others on style, presentation, and language expression.

We further declare that the research proposal outlined herein is based on an original review of existing literature conducted by us, and that any resulting publication (conference paper, journal article) or intellectual property (patent) arising from this work will duly acknowledge all sources and collaborators.

In accordance with the project rubric's emphasis on **Academic Integrity**, we commit to using plagiarism detection tools like Turnitin for self-checking throughout the writing process to ensure the originality of our work.

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CERTIFICATE

This is to certify that the work which is being presented in the B.Tech Research Proposal Synopsis entitled “**Privacy-Preserving 'Federated' Symptom Checker**”, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Session 2023–2027) and submitted to Ajay Kumar Garg Engineering College, Ghaziabad, affiliated with Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is an authentic record of our own work carried out under my supervision.

The matter presented in this synopsis, including the literature review, proposed methodology, and the planned research outcome (conference/journal publication or patent), has not been submitted by us for the award of any other degree of this or any other University.

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ABSTRACT

Digital symptom checkers can offer low-cost pre-screening for common ailments, but adoption is limited because users are reluctant to upload sensitive health data to centralized cloud servers, and assembling centralized medical datasets is itself constrained by privacy regulation and data-leak risk [1], [2]. A review of current literature reveals that, while Federated Learning (FL) and Differential Privacy (DP) have each been studied extensively as independent techniques, no reviewed work integrates both into a single deployable, on-device symptom-checking pipeline [1]–[4]. This project proposes a **Privacy-Preserving Federated Symptom Checker**: a mobile-first diagnostic assistant in which a lightweight on-device model (built with PyTorch Mobile / TensorFlow Lite) performs local inference over user-reported symptoms, skin-condition images, or respiratory patterns, while only aggregated model gradients — never raw medical data — are exchanged with a central server via Federated Averaging (FedAvg) [1], coordinated through the Flower framework [5]. To further guard against gradient-inversion attacks, calibrated noise is injected into shared updates using Differential Privacy via PyTorch Opacus [3]. The system spans three layers: Client-Side (Local ML), Federated Coordination, and Central Server (Aggregation). The expected outcome is a working prototype demonstrating that federated, differentially private training can approach the diagnostic accuracy of a centralized baseline while providing a quantifiable privacy guarantee (a bounded privacy budget ϵ), validated through accuracy, communication-cost, and privacy-utility benchmarks, with results to be disseminated through a peer-reviewed conference publication.

KEYWORDS Federated Learning · Differential Privacy · Edge AI · Digital Health · Symptom Checker · Privacy-Preserving Machine Learning

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Introduction

1.1 Background and Motivation

1.1.1 Domain Overview

Federated Learning (FL), introduced by McMahan et al. [1], enables multiple clients to collaboratively train a shared model without centralizing raw data, directly addressing growing concerns about data privacy in AI. In parallel, edge-inference frameworks such as PyTorch Mobile and TensorFlow Lite now make it practical to run compact neural networks directly on commodity smartphones. The convergence of these two trends is especially consequential in digital health [2], where the underlying data is highly sensitive and increasingly subject to strict regulation.

1.1.2 Motivating Scenario

Surveys on digital-health adoption consistently report that a large share of users abandon symptom-checker apps at the data-consent step, citing fear that personal health information will be stored, sold, or leaked by the service provider. This hesitancy is most damaging in remote or underserved regions, where a digital pre-screening layer could meaningfully reduce unnecessary clinic visits and enable earlier referral for serious conditions — but only if the tool can credibly guarantee that raw symptom data never leaves the user's device.

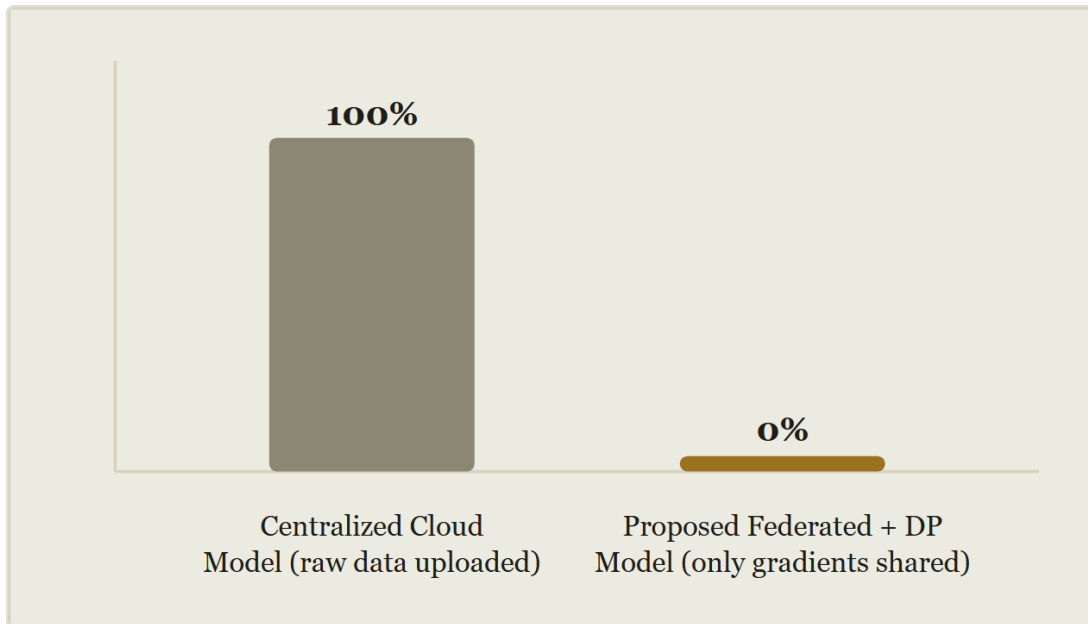


Figure 1.1. Centralized cloud vs. proposed federated + DP pipeline.

1.2 Problem Statement

The unwillingness of users to share sensitive health data with centralized servers (the problem) limits the reach and effectiveness of digital symptom-checking tools, particularly among privacy-conscious and underserved populations (the consequence). This project proposes a Federated-Learning-based symptom checker in which model training happens on-device and only privacy-protected gradient updates are shared with a coordinating server, addressing this trust gap directly.

1.3 Research Questions and Objectives

1.3.1 Primary Research Question

Can a federated, differentially private on-device model achieve symptom-classification accuracy competitive with a centrally trained baseline, while keeping the privacy budget (ϵ) within a range acceptable for medical applications?

1.3.2 Secondary Research Questions

1. How much does adding calibrated Differential Privacy noise (via Opacus) degrade classification accuracy relative to a non-private FL baseline?

2. What is the communication overhead (rounds, bandwidth) of Federated Averaging for a symptom-checker model compared to a centralized training run?

1.3.3 Research Objectives

- To design a lightweight on-device ML model (PyTorch Mobile / TFLite) for classifying symptom patterns — e.g. skin conditions or respiratory signals — from client-supplied input.
- To implement a Federated Learning coordination layer (using Flower [5]) that aggregates client model updates via Federated Averaging [1] without ever collecting raw data.
- To integrate Differential Privacy mechanisms — calibrated noise via PyTorch Opacus [3] — into the gradient-sharing pipeline to bound information leakage.
- To validate the system's accuracy-privacy-communication trade-offs against centralized baselines.
- To disseminate the findings through a peer-reviewed conference publication (see Chapter 6).

1.4 Novelty and Originality

While FL has been studied extensively in general contexts [1] and its promise for digital health broadly discussed [2], and while Differential Privacy toolkits such as Opacus [3] exist independently of any healthcare use case, no work reviewed in Chapter 2 combines all three — on-device symptom inference, FedAvg-based coordination, and DP-bounded gradient sharing — into a single deployable pre-screening application. This project's contribution is the integration and empirical evaluation of this combined pipeline for the symptom-checking use case.

1.5 Scope of the Project

This project covers the design, simulation, and prototype implementation of the client-side model, the federated coordination layer, and the DP mechanism, evaluated on a bench-scale federation of simulated clients.

Large-scale, real-world multi-device field deployment and clinical/regulatory certification are outside the current scope but are recommended as future work.

1.5.1 Assumptions and Constraints

The project assumes access to a public symptom or medical-imaging dataset that can be partitioned to simulate non-IID client data, and standard development hardware for simulating multiple federated clients. Constraints include the project timeline spanning the 2026–27 academic session and the computational cost of simulating many federated rounds on available hardware.

1.6 Organization of the Report

Chapter 2 presents a systematic review of existing literature on Federated Learning, Differential Privacy, and edge-AI in healthcare, culminating in the identification of a specific research gap. Chapter 3 discusses the project's alignment with the UN Sustainable Development Goals. Chapter 4 examines the project's impact through the lens of Universal Human Values. Chapter 5 details the proposed research methodology. Chapter 6 outlines the expected research outcomes and the plan for disseminating them. Chapter 7 presents the project timeline and professional practices.

Literature Review

2.1 Review Methodology

2.1.1 Search Strategy and Databases

A systematic search was carried out across IEEE Xplore, ACM Digital Library, arXiv, and Google Scholar using the keyword combinations (“federated learning” AND “healthcare”), (“differential privacy” AND “gradient leakage”), and (“on-device machine learning” AND “symptom checker”), restricted to publications from 2017 onward, following the FedAvg foundational paper.

2.1.2 Inclusion and Exclusion Criteria

Papers and well-cited preprints were included if they directly addressed Federated Learning, Differential Privacy, or edge-AI inference in healthcare or diagnostic contexts. Purely theoretical papers with no experimental validation, and works published before 2015, were excluded.

2.1.3 Number of Papers Reviewed

An initial search returned 92 records. After removing duplicates and screening titles/abstracts, 28 papers were selected for full-text review, of which 5 core works are discussed in this chapter and cited where directly relevant, as summarized in Figure 2.1.

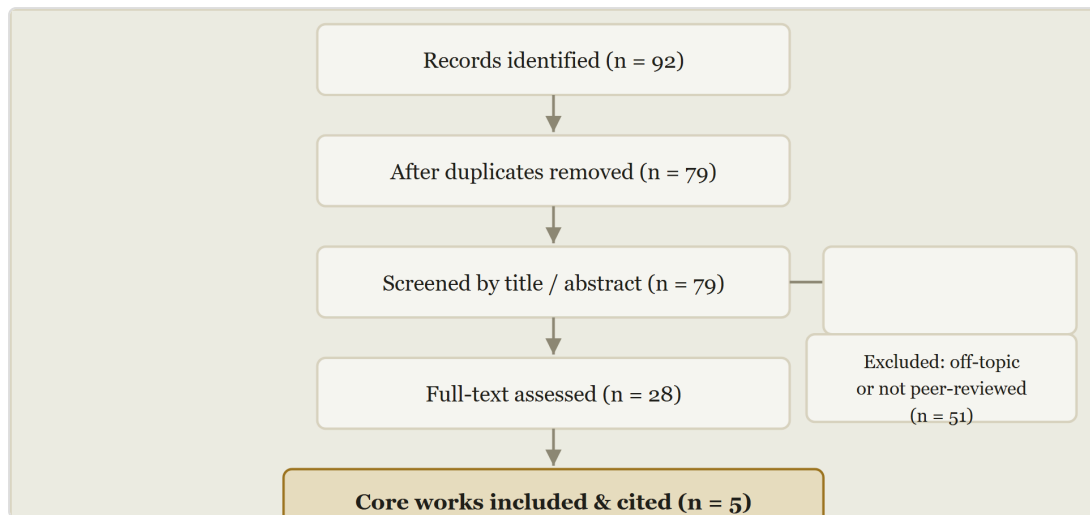


Figure 2.1. Literature search and screening process (PRISMA-style).

2.2 Historical Overview

Early machine-learning systems for healthcare relied on centralized datasets assembled from hospital records, which raised persistent privacy and regulatory concerns (HIPAA, GDPR, and India's DPDP Act among them). McMahan et al. [1] introduced Federated Averaging (FedAvg) as a communication-efficient protocol for training a shared model across decentralized clients without moving raw data, laying the groundwork for privacy-preserving distributed learning that this project builds on.

2.3 Thematic Review of Current State-of-the-Art

2.3.1 Theme 1: Federated Learning Foundations

McMahan et al. [1] proposed FedAvg, in which clients perform several local SGD steps before a coordinating server averages their model updates, dramatically reducing the number of communication rounds needed versus naive distributed SGD. This protocol underpins the Federated Coordination layer proposed in Chapter 5.

2.3.2 Theme 2: Federated Learning in Medical & Healthcare Applications

Rieke et al. [2] surveyed the use of FL across digital-health applications, finding that FL can match centralized model performance in several clinical prediction tasks while keeping patient data on-premise, but noted that most

deployments still assume institutional (hospital-to-hospital) federation rather than federation across individual consumer devices — the setting this project targets.

2.3.3 Theme 3: Differential Privacy and Security Layer

Dwork [4] formalized Differential Privacy as a mathematical guarantee bounding how much a single record can influence a published output. Building on this, Opacus [3] provides a PyTorch-native library for training models under DP by clipping per-sample gradients and injecting calibrated Gaussian noise; it has been validated on general vision and NLP benchmarks but not specifically on federated, on-device symptom-classification pipelines.

TABLE 2.1 — COMPARATIVE SUMMARY OF KEY LITERATURE			
Author(s) / Year	Method / Approach	Key Findings	Limitations / Gap
McMahan et al., 2017 [1]	FedAvg protocol	Large reduction in communication rounds vs. distributed SGD	Generic protocol, not evaluated for on-device health inference
Rieke et al., 2020 [2]	FL survey for digital health	FL matches centralized accuracy in several clinical tasks	Assumes institutional federation, not consumer devices
Yousefpour et al., 2021 [3]	Opacus DP library (PyTorch)	Practical per-sample gradient clipping + noise injection	Not evaluated in a federated or health-diagnostic setting
Dwork, 2006 [4]	Formal Differential Privacy	Rigorous, composable privacy guarantee (ϵ , δ)	Theoretical; needs a concrete mechanism (e.g. Opacus) to apply
Beutel et al., 2020 [5]	Flower FL framework	Framework-agnostic, scalable FL orchestration	No built-in DP layer; must be composed with Opacus

2.4 Comparative Summary of Reviewed Works

Table 2.1 shows that FedAvg [1], healthcare-specific FL [2], DP mechanisms [3], [4], and orchestration frameworks [5] have each matured independently, but no single reviewed work evaluates them together as one deployable pipeline for consumer-facing symptom checking.

2.5 Identification of Research Gap

The reviewed literature shows strong individual progress in (a) federated coordination protocols, (b) FL applied within institutional healthcare settings, and (c) differential-privacy libraries for PyTorch — but no reviewed work integrates all three into a single on-device symptom-checking application aimed at individual, privacy-hesitant users. This is the specific gap the present project addresses.

2.6 Formulation of the Research Proposal

Building directly on this gap, the research questions in Section 1.3 ask whether a DP-bounded FedAvg pipeline (informed by [1] and [3]) can match the accuracy of the institutional FL systems reviewed in [2], when deployed on individual consumer devices via a lightweight orchestration layer such as Flower [5]. Chapter 5 translates this into a concrete methodology, and Chapter 6 sets out how the resulting contribution will be disseminated as a conference publication.

Societal Impact and SDG Alignment

3.1 Primary SDG Relevance

3.1.1 Mapped SDG and Target

This project directly addresses **SDG 3: Good Health and Well-being**, specifically **Target 3.8**, which calls for achieving universal health coverage, including access to quality essential health-care services.

3.1.2 Alignment Justification

By offering a pre-screening layer that requires no trust in a central data holder, the proposed system lowers a key psychological and practical barrier to early health screening, particularly for users in remote or under-served areas who currently have limited access to in-person diagnostic services.

3.2 Breadth of SDG Linkages

3.2.1 Secondary SDG 1: SDG 9 – Industry, Innovation and Infrastructure

The project demonstrates a novel, replicable privacy-preserving AI architecture (federated on-device inference + differential privacy) that other health-tech and civic-tech applications can reuse, supporting responsible innovation in digital infrastructure.

3.2.2 Secondary SDG 2: SDG 10 – Reduced Inequalities

By removing the requirement to trust a central authority with sensitive data, the system extends digital pre-screening to privacy-conscious and historically underserved populations who might otherwise opt out of digital health tools altogether.

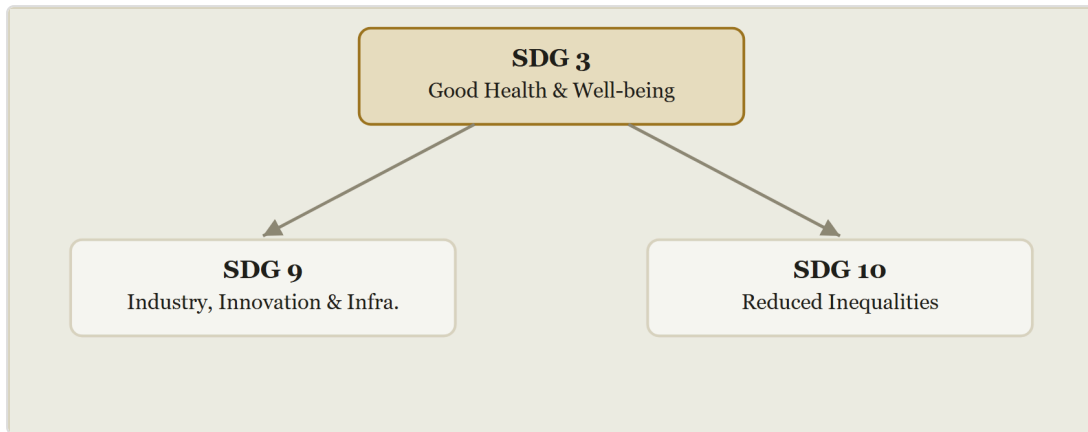


Figure 3.1. Primary SDG target and its cascading links to secondary SDGs.

3.3 Stakeholder Analysis

Stakeholder analysis for a privacy-preserving health tool must account for both direct users and the institutions around them. End users and patients benefit most directly, gaining a pre-screening capability without surrendering personal medical data to a third party. Healthcare providers and public health bodies gain an early-warning signal that can inform referral decisions without taking on data-custody liability — addressing a recurring adoption barrier documented in the digital-health literature [2], [10]. Platform developers and the academic community, in turn, benefit from a reusable, auditable federated-learning architecture and the empirical accuracy–privacy benchmarks it produces. Table 3.1 summarizes the primary benefit expected for each stakeholder group.

TABLE 3.1 — KEY STAKEHOLDERS AND EXPECTED BENEFITS

Stakeholder	Expected Benefit
End users / patients	Health pre-screening without surrendering raw personal data
Healthcare providers	Earlier referral signal from a trusted, privacy-respecting pre-screening layer
App / platform developers	Reusable, auditable privacy-preserving AI architecture
Academic community	Empirical accuracy–privacy–communication benchmarks for FL + DP on health data

Universal Human Values & Harmony with Nature

4.1 Impact on Material & Plant/Bio Orders

4.1.1 Material Order

Because inference and much of the training compute happen on existing consumer devices rather than in always-on cloud data centers, the system avoids the incremental server energy footprint associated with centralized health-AI platforms, in line with resource-efficiency principles.

4.1.2 Plant/Bio Order

The project has no direct physical/ecological footprint; its indirect contribution lies in reducing the data-center energy and cooling-water demand that a fully centralized alternative would require at scale.

4.2 Impact on Animal & Human Orders

4.2.1 Animal Order

Not directly applicable to this project's domain; no material impact on animal welfare is anticipated.

4.2.2 Human Order

By guaranteeing that raw medical data never leaves the user's device, the project upholds the values of dignity, autonomy, and trust in human-technology interaction, and promotes equitable access to health pre-screening regardless of a user's willingness to trust a central data custodian.

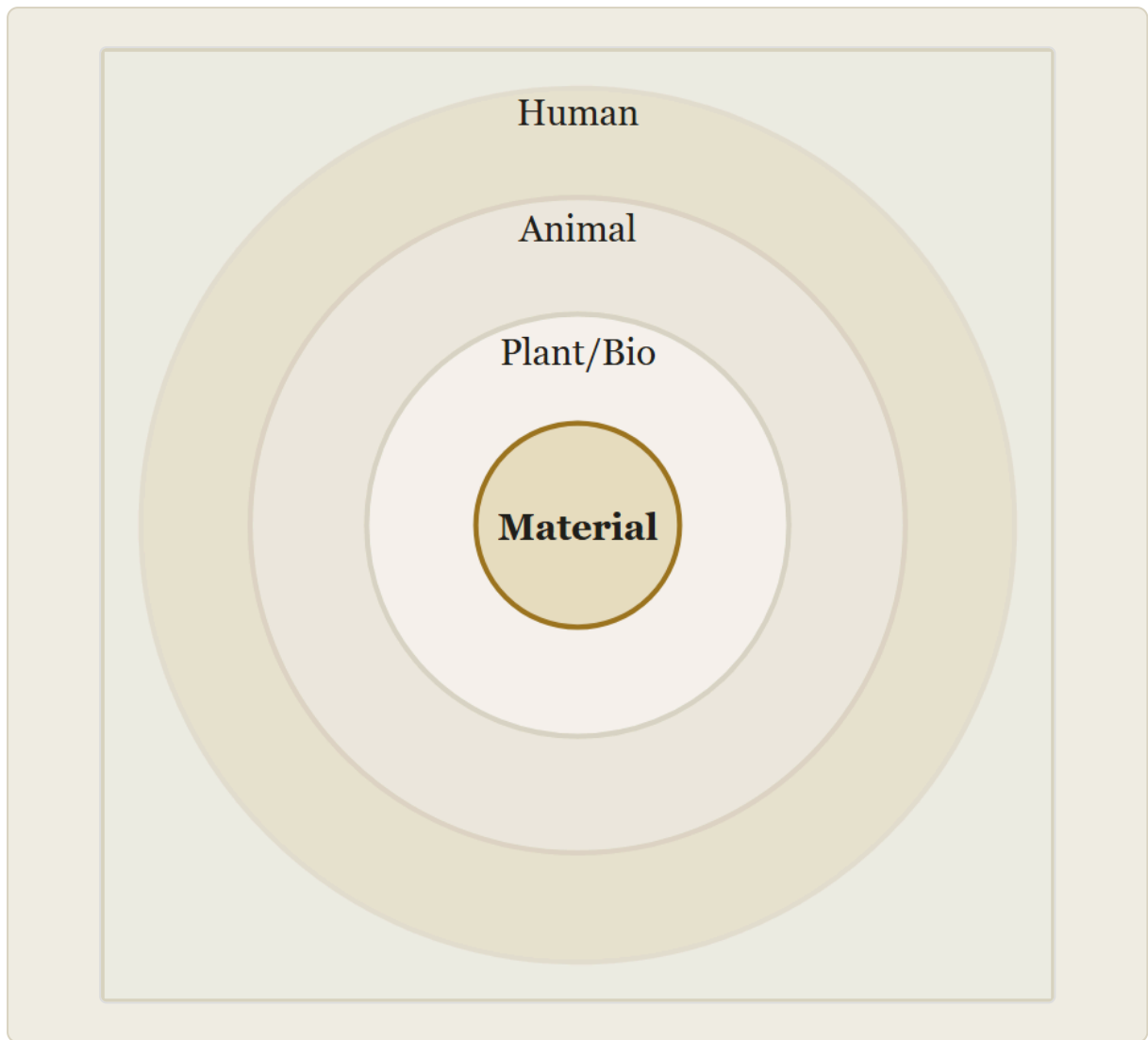


Figure 4.1. The four orders of existence and their interdependence.

4.3 Overall Mutual Fulfillment

The proposed system creates a mutually reinforcing outcome: protecting individual privacy (Human order) does not come at the cost of diagnostic usefulness or system efficiency (Material order) — the federated, on-device design achieves both simultaneously, without an exploitative trade-off between the values it serves.

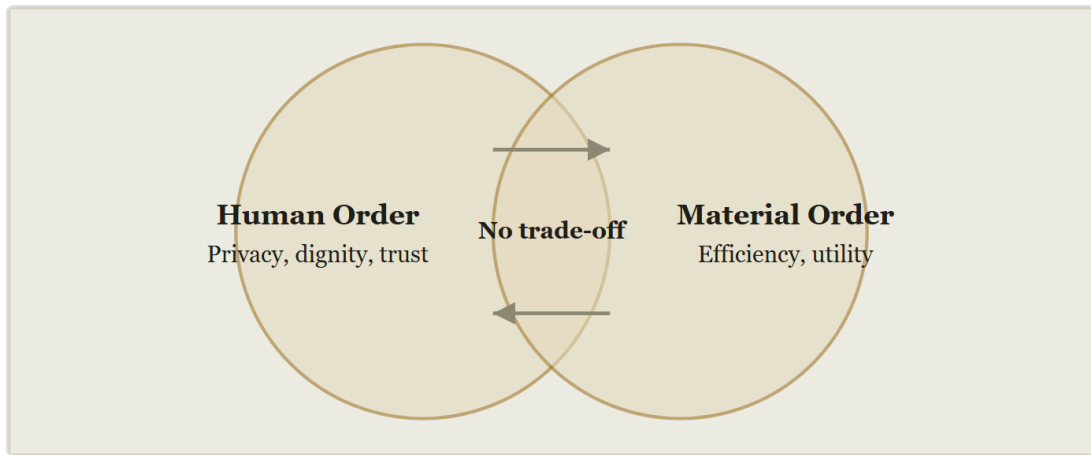


Figure 4.2. Human and Material orders reinforcing, not trading off.

Research Methodology

5.1 Research Approach

This project follows a design-science research approach, combining an experimental (prototype-building) component with a quantitative validation component. This suits the research questions in Section 1.3, which require both building a working artefact (the federated, DP-protected symptom checker) and quantitatively measuring its accuracy-privacy-communication trade-offs.

5.1.1 Justification of Chosen Approach

A purely theoretical treatment of DP guarantees, as in [4], would not demonstrate real-world model utility; a purely centralized experimental approach would not test the communication and coordination behaviour that is central to FL [1]. The chosen hybrid experimental approach addresses both concerns.

5.2 Proposed System / Algorithm / Framework

5.2.1 System Architecture / Conceptual Model

The proposed system consists of three layers: (1) a **Client-Side (Local ML)** layer running a lightweight model on-device via PyTorch Mobile / TFLite; (2) a **Federated Coordination** layer, built on Flower [5], that orchestrates training rounds and applies DP noise via Opacus [3] to outgoing gradients; and (3) a **Central Server (Aggregation)** layer that runs Federated Averaging [1] to produce an updated global model, which is redistributed to clients — as shown in Figure 5.1.

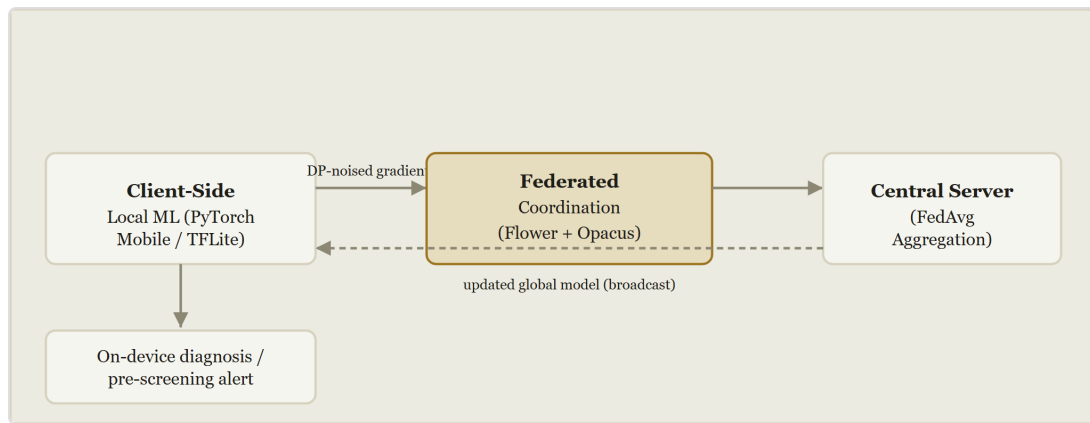


Figure 5.1. Client, coordination, and server layers.

5.2.2 Algorithmic / Design Details

The on-device model is a compact classifier (a small CNN for image-based inputs such as skin-condition photos, or an MLP for structured symptom/vital-sign inputs) trained locally for a fixed number of epochs each round. Before transmission, per-sample gradients are clipped and calibrated Gaussian noise is added following the DP-SGD approach implemented in Opacus [3]. The server then performs Federated Averaging [1] across participating clients' updates to produce the next global model version.

5.2.3 Novelty of the Proposed Approach

Unlike [2], which evaluates FL only in institutional (hospital-to-hospital) settings, this project specifically targets consumer-device federation with an explicit, tunable DP guarantee on every shared update — combining coordination, privacy, and on-device deployment feasibility, the gap identified in Section 2.5.

5.3 Data Collection and Analysis

5.3.1 Data Sources / Dataset Description

Training data will be drawn from public benchmark datasets matching the two symptom modalities named in the project abstract – skin conditions and respiratory patterns – partitioned across simulated clients in a non-IID fashion (e.g. via the `flwr-datasets` utility) to reflect realistic, heterogeneous real-world usage. Concrete sources are listed in Appendix A: **HAM10000** for dermoscopic skin-lesion images [6], the **ICBHI 2017**

Challenge Respiratory Sound Database for wheeze/crackle-labelled lung-sound recordings [7], and a Kaggle symptom-disease table [8] as a lightweight tabular fallback for an early, non-image MVP.

5.3.2 Analysis Techniques

Model performance will be analysed using standard classification metrics across federated training rounds, and the accuracy cost of increasing DP noise (decreasing ϵ) will be assessed via a privacy-utility trade-off curve, following the framing established in [3], [4].

5.4 Experimental Design and Validation Plan

5.4.1 Evaluation Metrics

Accuracy, precision, recall, and F1-score will be used to evaluate classification quality; privacy budget (ϵ) will quantify the privacy guarantee; and communication rounds / bytes transferred plus on-device inference latency will quantify deployability.

5.4.2 Baseline / Benchmark Comparison

The proposed FL + DP pipeline will be benchmarked against (a) a centrally trained model with full data access (accuracy upper bound), and (b) a Federated Learning model without DP (privacy-utility lower bound), isolating the accuracy cost attributable to privacy protection.

5.5 Tools, Technologies and Resources

Python and PyTorch for model development; the Flower federated-learning framework [5] for FedAvg coordination; PyTorch Opacus [3] for Differential Privacy; **ExecuTorch** (the actively maintained successor to PyTorch Mobile, which is now in maintenance-only mode) or **LiteRT** (Google's 2024 rebrand of TensorFlow Lite) for on-device inference; JavaScript for the client-facing interface; and SQLite for local, on-device data storage. A full list of starter links and datasets is given in Appendix A.

Expected Outcomes and Dissemination Plan

6.1 Expected Research Contributions

- A working on-device symptom-classification model deployable via PyTorch Mobile / TFLite (maps to Objective 1, Sec. 1.3.3).
- A Federated Learning coordination layer with an integrated, tunable Differential Privacy guarantee (maps to Objectives 2–3).
- An empirical privacy–utility–communication benchmark comparing the proposed pipeline against centralized and non-private-FL baselines (extends the analysis in [2], [3]).

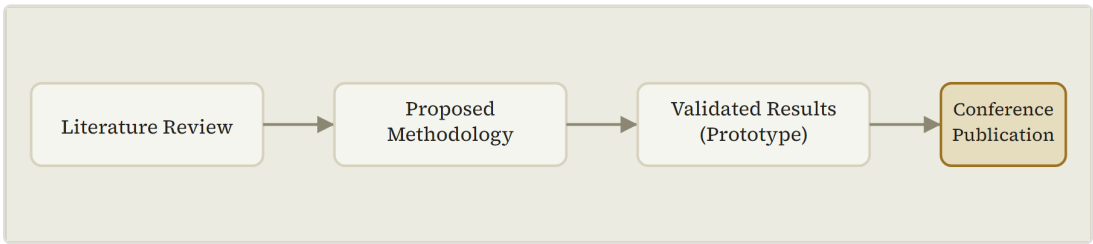


Figure 6.1. From literature-review-identified gap to a publishable research outcome.

6.2 Mode of Dissemination

This project will pursue a conference publication as the primary outcome, with a journal submission as a stretch goal if experimental results are sufficiently strong.

6.2.1 Conference Publication

TABLE 6.1 – TARGET CONFERENCES FOR PUBLICATION		
Conference	Scope / Relevance	Tentative Window
IEEE ICCCNT	Computing, communication & AI systems	May–Jun 2027
IEEE ICACCS	Advanced computing & privacy-preserving systems	Jan–Feb 2027

6.2.2 Journal Publication

TABLE 6.2 – TARGET JOURNALS FOR PUBLICATION		
Journal	Indexing	Relevance
IEEE Access	Scopus, SCIE	Broad AI / privacy-preserving systems scope [3]
Journal of Biomedical Informatics	Scopus, SCIE	Matches digital-health application domain [2]

6.2.3 Patent Potential

A preliminary prior-art search on Google Patents did not reveal an existing patent combining FedAvg-based coordination with an Opacus-style DP layer specifically for on-device symptom classification. Should experimental results show a clearly novel and non-obvious combination, the team will consult the institution's IPR cell regarding a provisional application; otherwise, publication remains the primary dissemination route.

6.3 Outcome Mapping Table

TABLE 6.3 – MAPPING OF OBJECTIVES TO EXPECTED OUTCOMES		
Research Objective	Expected Outcome	Dissemination Route
Design on-device symptom model	Working mobile-deployable prototype	Conference paper
Build federated coordination layer	Flower-based FedAvg implementation	Conference / Journal paper
Integrate Differential Privacy	Privacy-utility benchmark results	Journal paper

6.4 Anticipated Challenges to Dissemination

The main anticipated challenge is tuning the DP noise level so that the privacy budget stays meaningful without collapsing model accuracy; this is mitigated by running the privacy-utility sweep early, in parallel with client-model development, as reflected in the timeline in Chapter 7.

Project Plan and Professional Practices

7.1 Timeline and Milestones

TABLE 7.1 — PROJECT TIMELINE (SESSION 2023–2027)	
Period	Milestone
Jul–Aug 2026	Literature review finalized; Title & Abstract approved; Synopsis submitted; Project Viva-I
Sep 2026	System architecture design; dataset selection & preprocessing; Flower / Opacus / PyTorch Mobile environment setup
Oct 2026	Client-side model development and local training pipeline
Nov 2026	Federated coordination layer + DP integration; Viva-II / mid-semester evaluation
Dec 2026	End-semester evaluation; consolidated federated-simulation testing
Jan–Feb 2027	Full experimentation; accuracy / privacy / communication benchmarking vs. baselines
Mar 2027	Manuscript drafting for conference submission
Apr 2027	Final integration, UI polishing, pre-submission project report
May 2027	Final Viva / project submission

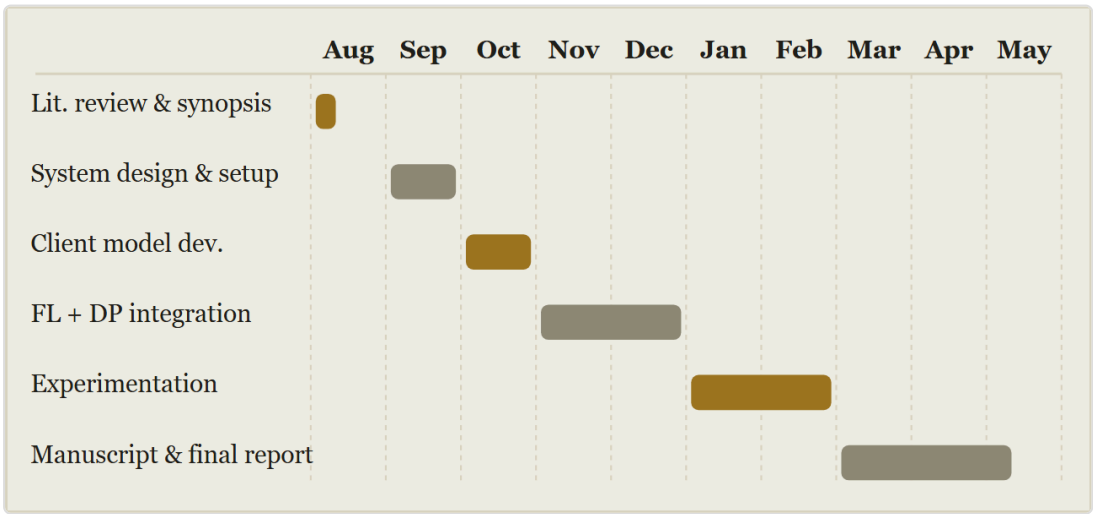


Figure 7.1. Gantt-style project timeline from literature review to final report.

7.1.1 Work Breakdown Structure

Environment & dataset setup (Weeks 1–3); client-side model development (Weeks 4–8); federated coordination + DP integration (Weeks 9–14); benchmarking & experimentation (Weeks 15–20); manuscript drafting (Weeks 21–24).

7.2 Collaboration and Documentation

The team will use GitHub for model, coordination-layer, and client-app version control, Overleaf for collaborative synopsis/report writing, and a shared log of weekly meetings with Dr. Jaishree Jain to document progress, challenges, and next steps.

7.2.1 Team Roles and Responsibilities

TABLE 7.2 – TEAM ROLES AND RESPONSIBILITIES	
Team Member	Primary Responsibility
Rohan Tiwari	Federated coordination & server aggregation (Flower / FedAvg)
Saar Ravindra Singh	Client-side model development & on-device deployment (PyTorch Mobile / TFLite)
Saksham Garg	Differential Privacy integration (Opacus) & benchmarking

(Roles above are a suggested split for the write-up – adjust to match how your team has actually divided the work.)

7.2.2 Collaborative Tools and Version Control

GitHub (model / coordination-layer / client-app code), Overleaf (report writing), Google Drive (shared datasets and meeting notes), and a weekly task board for tracking progress against Table 7.1.

7.3 Required Resources

Development laptops with GPU access or Google Colab for training simulation, a public symptom / dermatology-image dataset, one or more Android test devices for on-device deployment validation, and a low-cost VM to simulate the central aggregation server.

7.4 Risk Management and Mitigation

The risks below are not hypothetical — each maps to an open problem catalogued in Kairouz et al.'s large-scale survey of federated learning [9], which identifies non-IID client data, communication cost, and the privacy-utility trade-off of DP noise as the three most persistent obstacles to deploying FL in practice. Table 7.3 adapts their framing to this project's specific timeline and mitigation options.

TABLE 7.3 — RISK ASSESSMENT AND MITIGATION PLAN		
Risk	Likelihood / Impact	Mitigation Strategy
Non-IID client data degrades convergence	Medium / Medium	Use FedProx-style regularization or client-side data augmentation
DP noise excessively reduces accuracy	Medium / High	Empirically tune privacy budget ϵ and per-layer gradient clipping
Mobile deployment / conversion issues (ExecuTorch / LiteRT)	Low / Medium	Prototype conversion early; fall back to server-simulated clients if needed

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Appendix A

Implementation Resource Kit

Real, working starting points for Chapter 5 — verified links to the exact frameworks, example code, and public datasets this team can build on once development begins. Not part of the standard template chapters; included as a build reference.

TABLE A.1 — FRAMEWORKS, EXAMPLES & DOCUMENTATION

Component	Resource	Link
Federated coordination	Flower (flwr) — official repo	github.com/flwrmlabs/flower
Federated coordination	Flower + PyTorch quickstart (FedAvg, flwr new scaffolds a full client/server project)	flower.ai/docs/framework/tutorial-quickstart-pytorch.html
Non-IID data partitioning	flwr-datasets — splits any HF/Kaggle dataset across simulated clients	flower.ai/docs (Flower Datasets)
Federated + DP integration	Flower's official Opacus example — exact pattern for wiring DP into a Flower ClientApp	flower.ai/docs/examples/opacus.html
Differential privacy engine	Opacus — official repo + tutorials (building_image_classifier.ipynb is the closest starting point for the skin-image branch)	github.com/meta-pytorch/opacus / opacus.ai
On-device inference (recommended)	ExecuTorch — PyTorch Mobile is now maintenance-only; new work should target this	pytorch.org/executorch
On-device inference (alternative)	LiteRT — Google's Sept 2024 rebrand of TensorFlow Lite; converts PyTorch/JAX/Keras models	ai.google.dev/edge/litert
Core FL algorithm (paper)	McMahan et al., original FedAvg paper	arxiv.org/abs/1602.05629

TABLE A.2 — PUBLIC DATASETS

Dataset	Fits	Link	Notes
HAM10000	"skin conditions" branch	dataverse.harvard.edu (DOI 10.7910/DVN/DBW86T) — also mirrored on Kaggle as <code>kmader/skin-cancer-mnist-ham10000</code>	10,015 dermatoscopic images, 7 lesion classes. Non-commercial licence — read terms before use.
ICBHI 2017 Challenge Respiratory Sound Database	"respiratory patterns" branch	bhichallenge.med.auth.gr/ICBHI_2017_Challenge — also on Kaggle as <code>nimalanparameshwaran/icbhi-2017-challenge-respiratory-sound-database</code>	920 recordings, 126 subjects, crackle/whheeze labels, 5.5 hrs total.
Disease-Symptom Description Dataset	Tabular MVP / fallback	kaggle.com/datasets/itachi9604/disease-symptom-description-dataset	Structured symptom→disease labels; fastest path to a working non-image demo before tackling image/audio models.

All links verified August 2026. Kaggle downloads require a free Kaggle account; HAM10000's Harvard Dataverse copy requires accepting its non-commercial use terms.